

1. Mathematical Model.

Membrane Voltage: $\leftarrow$ post synaptic neuron

$$\frac{dV}{dt} = -\pi + I(t)$$

Calcium Variable: $\leftarrow$ post synaptic neuron

$$\frac{dC(t)}{dt} = -\frac{1}{\tau_c} C(t) + J_c \sum_i \delta(t-t_i)$$

Synaptic Variable:

(1) with spike:

$$X \rightarrow X+a \text{ if } V(t_{pre}) > \theta_V \text{ and } \theta_{up}^l < C(t_{pre}) < \theta_{up}^h$$

$$X \rightarrow X-b \text{ if } V(t_{pre}) \leq \theta_V \text{ and } \theta_{down}^l < C(t_{pre}) < \theta_{down}^h$$

(2) without spike:

$$\frac{dx}{dt} = \alpha \quad \text{if } x > \theta_x$$

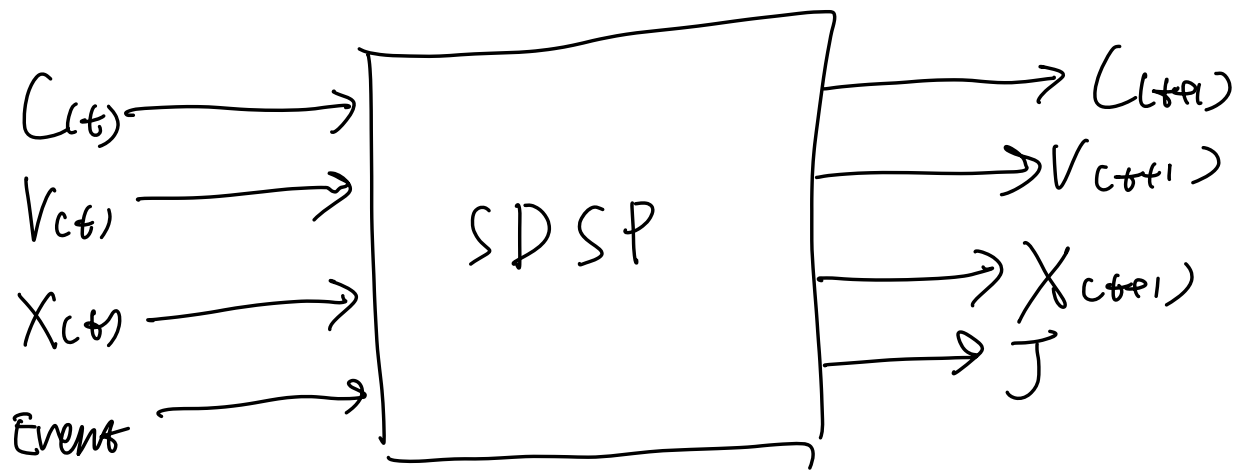
$$\frac{dx}{dt} = -\beta \quad \text{if } x \leq \theta_x$$

Binary Weight J :

$$J = J_+ \quad \text{if } X(t_{pre}) > \theta_x$$

$$J = J_- \quad \text{if } X(t_{pre}) \leq \theta_x$$

Arch



V_m : can be extracted directly from neuron model .

$C(t)$: $\sim$ can be regarded as a pulse counter

ODZIV

$\begin{matrix} X \\ J \end{matrix} \} \rightarrow W \text{ (3-bit)}$

$C \rightarrow$ 3-bit counter

if spike, counter ++

el if time ref, counter --

(configurable F)

$\begin{matrix} V_{mem} \\ C \end{matrix} \} \Rightarrow \begin{cases} V_{up} & \text{if } V_{mem} \geq \theta_m, \theta_1 \leq Ca < \theta_3 \\ V_{down} & \text{if } V_{mem} < \theta_m, \theta_1 \leq Ca < \theta_2 \end{cases}$

$\tau_c \rightarrow$ time ref (continuous?)

$W \rightarrow W+1$ if $V_{up}=1$ and $spike=true$

$W \rightarrow W-1$ if $V_{down}=1$ and $spike=true$

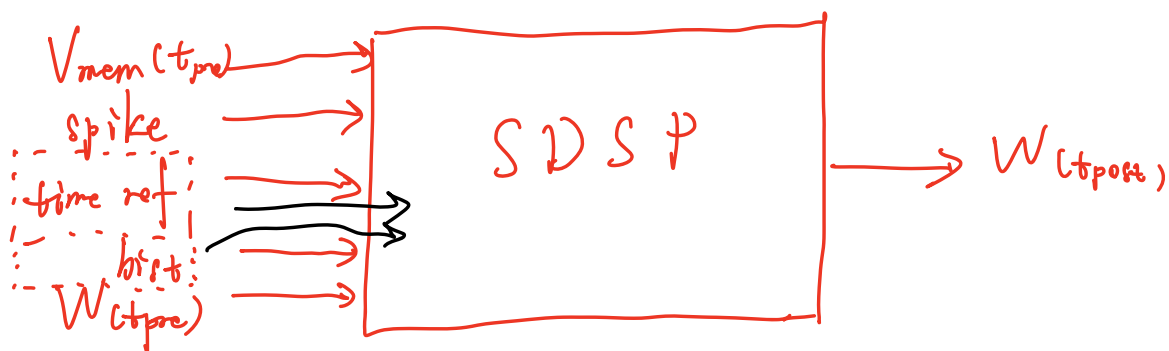
W : 3-bit

drift $\rightarrow$ bistability The highest bit of $\Rightarrow W[2]$

$W \rightarrow W+1$ if $W[2]=1$ and $bist=true$

$W \rightarrow W-1$ if $W[2]=0$ and $bist=true$

Interfaces of SDSP block



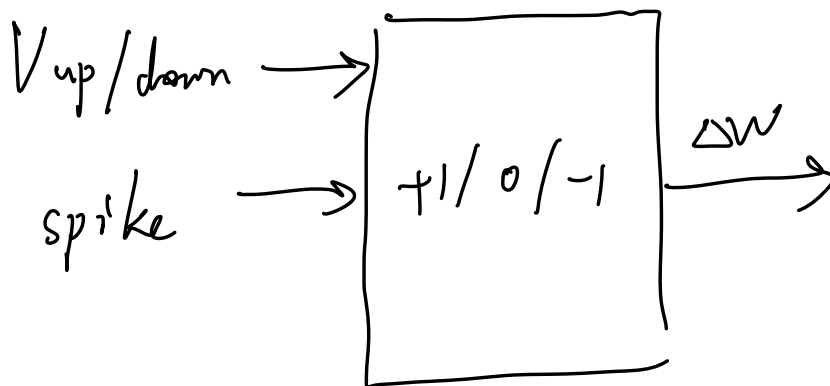
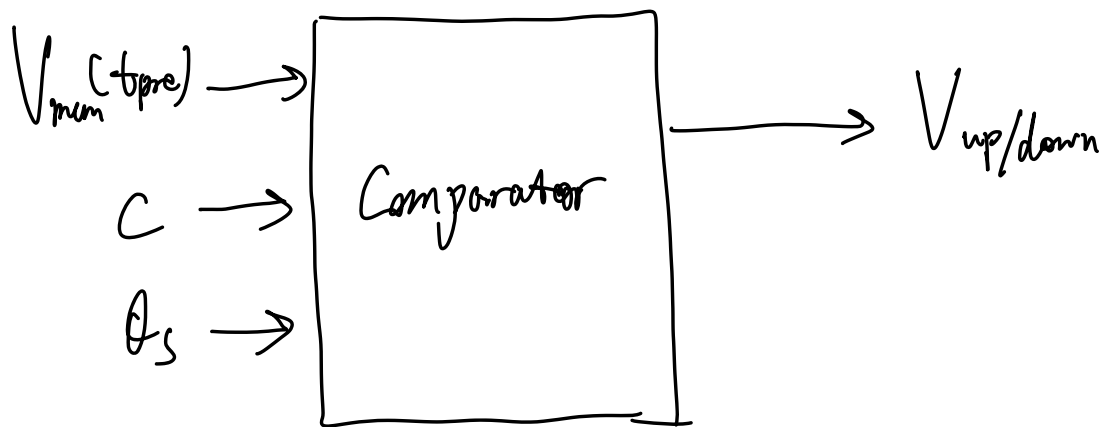
time ref and bist can be kept inside of SDSP block

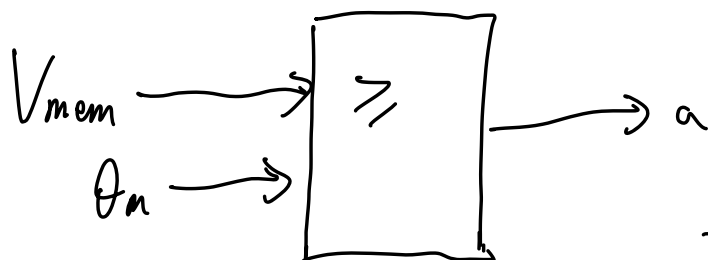
$V_{mem}(t_{pre}) \leftarrow$ post neuron

spike $\leftarrow$ pre neuron

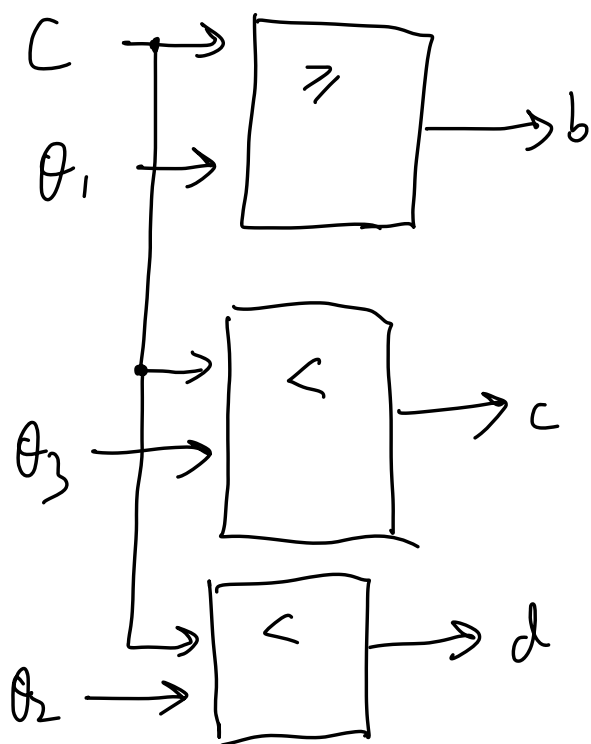
W from some memory blocks.

Comparator (truth table?)



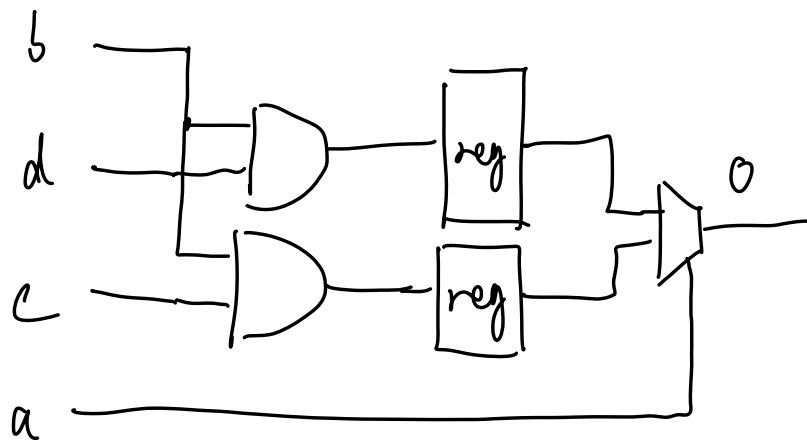


a	b	c	d	up	down
0	0	0	0	x	0
0	0	0	1	x	0
0	0	1	0	x	0
0	0	1	1	x	0
0	1	0	0	x	0
0	1	0	1	x	1
0	1	1	0	x	0
0	1	1	1	x	1
1	0	0	0	0	x
1	0	0	1	0	x
1	0	1	0	0	x
1	0	1	1	0	x
1	1	0	0	0	x
1	1	0	1	0	x
1	1	1	0	1	x
1	1	1	1	1	x



b	c	up
0	0	0
0	1	0
1	0	0
1	1	1

b	d	down
0	0	0
0	1	0
1	0	0
1	1	1



a	0	ΔW
0	0	0
0	1	-1
1	0	0
1	1	1

Bistability

